INCH-POUND

MIL-M-14/2
2 April 1990

MILITARY SPECIFICATION SHEET

MOLDING COMPOUNDS, PHENOLIC, THERMOSETTING, CONTAINING MINERAL/GLASS FILLERS

This specification is approved for use by all departments and agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-M-14.

Phenolic resin molding compounds, mineral/glass filled, shall be of the following types, as specified:

Type MFE	–	Mineral filler, electrical properties
Type MFH	–	Mineral filler, heat-resistant
Type GPG	–	Glass filler, general purpose
Type GPI-5	–	Glass filler, impact-resistant; nominal impact strength, 0.5 foot-pound per inch notch
Type GPI-10	–	Glass filler, impact-resistant; nominal impact strength, 1.0 foot-pound per inch notch
Type GPI-20	–	Glass filler, impact-resistant; nominal impact strength, 2.0 foot-pounds per inch notch
Type GPI-30	–	Glass filler, impact-resistant; nominal impact strength, 3.0 foot-pounds per inch notch
Type GPI-50	–	Glass filler, impact-resistant; nominal impact strength, 5.0 foot-pounds per inch notch
Type GPI-100	–	Glass fiber filler, impact-resistant; nominal impact strength, 10.0 foot-pounds per inch notch.

AMSC N/A FSC 9330

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REQUIREMENTS

Qualification test requirements are specified in table I. Batch acceptant test requirements are specified in table II.

NOTES

Type MFE. This type is a low-loss, high-dielectric-strength, low-water absorption mineral-filled phenolic compound.

Type MFH. This type is a mineral-filled phenolic compound intended for applications requiring heat resistance.

Type GPG. This type is a general purpose glass-filled phenolic compound intended for applications requiring good mechanical, electrical, and heat resistant properties.

Type GPI-5. This type is a heat resistant, moderate-impact, glass-filled phenolic compound having good electrical properties.

Type GPI-10. This type is a heat-resistant, medium-impact, glass-filled phenolic compound having good electrical properties.

Type GPI-20. This type is a heat-resistant, of moderately high-impact, glass-filled phenolic compound having good electrical properties.

Type GPI-30. This type is a heat-resistant, high-impact, glass-filled phenolic compound having good electrical properties.

Type GPI-50. This type is a heat-resistant, high-impact, glass-filled phenolic compound having good electrical properties.

Type GPI-100. This type is a glass-fiber-filled phenolic resin molding compound of very high-impact strength and good electrical properties.

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TABLE I. *Qualification test requirements for phenolic resin molding compounds, mineral/glass filled.*

Requirement	Type MFE	Type MFH	Type GPG	Type GPI-5	Type GPI-10	Type GPI-20	Type GPI-30	Type GPI-50	Type GPI-100
Mechanical/physical									
Compressive strength, endwise	15,000	15,000	23,000	25,000	25,000	23,000	20,000	20,000	20,000
Flexural strength	8,000	7,000	9,000	12,000	12,000	12,000	12,000	14,000	15,000
Heat deflection temperature	115	130	170	175	175	175	175	175	200
Heat resistance	200	200	180	175	250	250	250	250	250
Impact strength, side[1]	–	0.25	0.3	0.5	1.0	2.0	3.0	5.0	10.0
Tensile strength	4,200	4,200	4,500	7,000	6,500	6,500	6,500	6,000	4,500
Water absorption	0.10	0.35	0.30	0.35	0.35	0.40	0.50	1.0	1.5
Electrical									
Dielectric breakdown:									
Short-time test (To be recorded as basis for determining initial voltage in step-by-step test)									
Step-by-step test	45	35	35	35	35	35	35	35	40
Short-time test (To be recorded as basis for determining initial voltage in step-by-step test)									
Step-by-step test	40	10	15	15	15	15	15	15	15
Dielectric constant:									
at 1 kilohertz	6.0	–	6.5	6.0	6.0	6.0	6.0	6.0	6.0
	6.0	–	7.0	7.0	7.0	7.0	7.0	7.0	7.0
at 1 megahertz	6.0	–	5.7	5.7	5.7	5.7	5.7	5.7	5.7
	6.0	–	6.0	6.0	6.0	6.0	6.0	6.0	6.0

See footnote at end of table

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TABLE I. *Qualification test requirements for phenolic resin molding compounds, mineral/glass filled – Continued.*

Requirement	Type MFE	Type MFH	Type GPG	Type GPI-5	Type GPI-10	Type GPI-20	Type GPI-30	Type GPI-50	Type GPI-100
Electrical *(Continued)*									
Dielectric strength:									
Short-time test	325	215	275	250	250	250	250	250	300
Step-by-step test	275	150	225	200	200	200	200	200	200
Short-time test	325	125	225	175	175	175	175	175	75
Step-by-step test	275	80	200	125	125	125	125	125	50
Dissipation factor:									
at 1 kilohertz	0.03	–	0.04	0.04	0.05	0.04	0.04	0.04	0.04
	0.03		0.05	0.06	0.05	0.05	0.05	0.05	0.05
at 1 megahertz	0.15	–	0.03	0.03	0.03	0.03	0.03	0.03	0.03
	0.02		0.04	0.04	0.04	0.04	0.04	0.04	0.04
Surface resistance	5.0	–	–	–	–	–	–	–	–
Volume resistance	2.0	–	–	–	–	–	–	–	–
Combustion									
Flame resistance: *Units* – Seconds (min. avg.) *Test method* – D 229 See 4.5.2.9									
Ignition time	60	120	150	150	150	150	150	150	120
Burning time	210	150	150	150	150	150	150	150	120
Flammability: *Test method* – UL 94 See 4.5.2.10									
	V-0	V-0	V-0	V-0	V-0	V-0	V-0	V-0	V-0

[1]The side of the test specimen is that area formed by the chase of the mold.

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TABLE II. *Batch acceptance test requirements for phenolic resin molding compounds, mineral/glass filled.*

Property to be tested	Type MFE	Type MFH	Type GPG	Type GPI-5	Type GPI-10	Type GPI-20	Type GPI-30	Type GPI-50	Type GPI-100
Dielectric constant at 1 megahertz	6.0	–	6.0	6.0	6.0	6.0	6.0	6.0	6.0
Dielectric strength, step-by-step	275	80	200	125	125	125	125	125	50
Dissipation factor at 1 megahertz	0.02	–	0.04	0.04	0.04	0.04	0.04	0.04	0.04
Flexural strength	8,000	7,000	9,000	12,000	12,000	12,000	12,000	14,000	15,000
Impact strength, side[1]	–	0.25	0.3	0.5	1.0	2.0	3.0	5.0	10.0
Water absorption	0.10	0.35	0.30	0.35	0.35	0.40	0.50	1.0	1.5

[1]The side of the test specimen is that area formed by the chase of the mold.

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Preparing activity:
Navy - SH
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